

INSTRUCTIONAL DESIGN CASE STUDY

Think Before You Click:

Spotting Phishing Emails

A story-driven cybersecurity awareness module | Articulate Storyline 360

Developer	Trina
Tool	Articulate Storyline 360 (Trial)
Audience	New corporate employees — onboarding and annual compliance training
Slides	13 slides approximately 15–20 minutes
Completed	July 2026

1. Analysis

Business Problem

Phishing remains the most common entry point for corporate data breaches and ransomware incidents. A single employee clicking a malicious link or entering credentials into a spoofed login page can compromise an entire organisation's network, leading to financial loss, regulatory exposure, and reputational damage. Most breaches of this kind exploit human behaviour rather than technical vulnerabilities — meaning the fastest, lowest-cost mitigation is better employee judgement, not new software.

Training Need

New employees typically receive IT security policies during onboarding but little practical, scenario-based practice in recognising phishing attempts. There is a gap between employees knowing that phishing exists and employees being able to reliably identify a phishing email in the flow of their actual inbox, under time pressure, without a security expert looking over their shoulder.

Target Audience

- New corporate employees across departments (not IT specialists) completing onboarding or annual compliance training.

- Mixed age range and digital literacy levels; comfortable with email and basic office software.
- Likely completing this training on a laptop or desktop during work hours, possibly in a single sitting.
- Motivated by job security and organisational compliance requirements rather than personal interest in cybersecurity.

Prior Knowledge

Learners are assumed to have little or no formal cybersecurity training. They know how to use email and recognise the general idea of "scam messages," but they cannot reliably name the specific red flags of a phishing email (sender spoofing, urgency language, mismatched links, generic greetings) or explain the correct reporting procedure.

Learning Gap

The gap is not awareness (most employees know phishing is "bad") but applied recognition and correct action: knowing exactly what to look for in a real email, and knowing the safest next step — rather than guessing, ignoring the email, or clicking to "see what it is."

Constraints

- Development tool: Articulate Storyline 360, 30-day trial version — favours simpler interaction types over advanced variable-driven logic.
- Developer is a beginner in Storyline; build time is capped at roughly 8–10 hours.
- Module must be short enough for a single onboarding session: target 15–20 minutes of learner time.
- No dedicated voice-over budget assumed; narration scripts are provided but narration is optional (on-screen text must stand alone).
- Must publish cleanly to Web (HTML5) for portfolio hosting and direct browser access — no LMS/SCORM dependency required.

Delivery Method

Self-paced, asynchronous eLearning module published to Web (HTML5) and accessed directly via browser link, consistent with the other two portfolio projects.

Desktop/laptop-first design, since this is the primary context for new-hire onboarding.

Success Criteria

- 90% or more of learners score 80% or higher on the final assessment.
- Learners can correctly classify at least 3 of 4 sample emails as Safe, Suspicious, or Phishing in the scenario activity.
- Reduction in real-world phishing simulation click rates within 60–90 days of training rollout (measured outside the module, via the organisation's phishing-simulation tool).
- Increase in the number of suspicious emails correctly reported through official channels.

2. Design

SMART Learning Objectives

By the end of this module, learners will be able to:

- Identify at least four common warning signs of a phishing email (sender address, urgency language, suspicious links, generic greetings) with 100% of signs covered in the content and reinforced in practice.
- Correctly classify a set of sample emails as Safe, Suspicious, or Phishing with at least 80% accuracy during the scenario activity.
- Select the safest next action (do not click, verify sender, report) for a suspicious or phishing email in at least 4 of 5 scenario decisions.
- Describe the correct internal procedure for reporting a phishing email, demonstrated by scoring 100% on the reporting-procedure assessment question.

Instructional Strategy

The module follows a story-driven, problem-centred structure grounded in adult learning principles (Knowles' andragogy):

- Relevance first: the module opens with a relatable workplace scenario (a colleague nearly falling for a phishing email) rather than a definitions slide, so learners immediately see why this matters to their own job.

- Practice before polish: content slides are kept brief and are immediately followed by an interactive check, so learners apply each concept right after meeting it rather than absorbing a long lecture first.
- Realistic materials: all example emails, sender addresses, and scenarios mirror real corporate phishing patterns (fake IT alerts, fake invoices, fake HR notices) rather than generic textbook examples.
- Decision-making over recall: the centerpiece activity asks learners to decide what they would do, not just what they know, mirroring the real on-the-job moment of judgement.
- Low-stakes failure: incorrect answers throughout the module trigger explanatory feedback rather than penalty, encouraging exploration and reducing anxiety, consistent with adult learners' preference for self-directed, mistake-tolerant learning.

Assessment Strategy

- Formative checks are embedded throughout (a True/False knowledge check and the inbox scenario activity) and are low-stakes — used for practice, not scored toward completion.
- A final graded assessment of 5–7 questions (multiple choice, true/false, and one scenario-style question) is used to confirm mastery, with the results shown directly on-screen at course completion.
- Passing score is set at 80%, aligned to the module's stated success criteria.
- Every question — formative and summative — includes specific, explanatory feedback for both correct and incorrect responses, so the assessment itself continues to teach.

Interaction Strategy

Interaction types were deliberately chosen to be achievable within a 30-day trial and a beginner's skill level, while still requiring active decision-making rather than passive clicking. The module uses:

- Click-to-reveal panels for introducing concepts (the "anatomy of a phishing email" and the reporting steps).
- A tabs interaction for the four common phishing tactics.

- Hover states for a lightweight comparison of legitimate vs phishing email features.
- A simple two-option drag-and-drop matching activity for reinforcing terminology.
- A branching-style scenario built from button clicks and layers (no variables required beyond simple visited-state tracking) — the inbox activity.
- Storyline's built-in True/False and Multiple Choice graded question slides for the final assessment, which avoids any custom variable logic.

Visual Style Recommendations

- Clean, corporate-modern aesthetic consistent with a security/compliance topic: minimal illustration, confident use of whitespace, icon-led rather than image-heavy.
- A recurring "inbox" visual motif (browser-style window chrome, email card layout) ties the whole module together and keeps the scenario realistic.
- Consistent colour-coding for decision states: green for Safe, amber/gold for Suspicious, red for Phishing — used consistently across the scenario, feedback layers, and states so learners build a reliable visual shorthand.
- Flat-style icon set (e.g., from The Noun Project, Flaticon, or Storyline's built-in icon library) for envelope, shield, warning triangle, magnifying glass, and link icons.

Colour Palette

Colour	Hex	Use
Deep Navy	#1B3A5C	Headers, navigation bar, primary text on light backgrounds
Teal	#2E8B92	Accent colour, buttons, active states, tab highlights
Off-White	#FAF9F6	Slide backgrounds
Safe Green	#3D7A4E	"Safe" tag, correct feedback, positive states
Caution Gold	#C9A24B	"Suspicious" tag, warning accents
Alert Red	#B23A48	"Phishing" tag, incorrect feedback, critical warnings

Neutral Grey #595959 Secondary text, captions, instructions

Font Recommendations

- Headings: Open Sans Bold, 28–36pt — clean, modern, screen-friendly.
- Body text: Open Sans Regular, 18–20pt — high readability at a glance, appropriate for a workplace compliance module read on a laptop screen.
- Email-mockup text: Open Sans Regular for the simulated inbox slides, kept visually distinct through the browser-chrome window styling rather than a separate font family — one consistent typeface throughout keeps the module simple to build and maintain.
- Minimum body text size 18pt; avoid more than two font families in the entire module.

Accessibility Considerations

- All interactive objects (buttons, tabs, hotspots) are set as focus order objects with visible focus states and are reachable via keyboard/Tab navigation.
- Colour is never the only signal: Safe/Suspicious/Phishing tags use colour plus an icon plus a text label, so colour-blind learners are not disadvantaged.
- All narration scripts double as on-screen text or closed captions; the module is fully usable with audio off.
- Alt text is written for every meaningful image and icon; decorative graphics are marked decorative in Storyline's accessibility settings.
- Text-background contrast meets at least WCAG AA (4.5:1) throughout, including inside coloured feedback layers.
- Player is set to allow seeking/replay of narration and does not lock navigation in a way that traps keyboard-only users.

4. Implementation

This module is designed for deployment as a published Web (HTML5) output, hosted and accessed directly via browser link — consistent with the other two portfolio projects (Google Classroom and Keep Your Cool) — rather than through an LMS/SCORM package.

Publishing Settings

- Publish to Web from Storyline 360 (File > Publish > Web).
- Player: enable seekbar and volume control; keep Next/Back/Menu visible except on the Welcome, Assessment transition, and Completion slides as noted in the storyboard.
- Enable "Restrict learner navigation" only to the extent of preventing skipping the final assessment (i.e., learners can freely browse content slides, but the assessment must be reached via the normal course flow).
- Set player size to responsive/fixed 720p (1280×720) for consistent display on standard laptop screens.
- Enable closed captions toggle in the player if narration is recorded, using the scripts provided in Section 3 as caption source text.

Completion Experience

- Since the module is published to Web rather than an LMS, there is no SCORM completion/pass-score reporting; the final assessment score and pass/fail result are shown directly to the learner on-screen at the Summary & Completion slide.
- Passing score: 80% (matches the Design section's assessment strategy), displayed on the results screen with pass/retry messaging.
- Completion: the module closes via an Exit Course trigger on the final Summary & Completion slide (Slide 13), consistent with how the other two Storyline portfolio projects close out.
- Retake policy: learners who score below 80% are shown a "Review and Retry" path (Storyline's built-in quiz result slide, with a "Retry Quiz" button) rather than being locked out, consistent with the module's low-stakes, mistake-tolerant design philosophy.

5. Evaluation — Kirkpatrick Model

The following framework outlines how the effectiveness of this training would be evaluated after rollout, using all four levels of Kirkpatrick's model.

Level 1: Reaction	Post-course rating survey (1–5 scale) embedded as the final course screen, asking whether the content felt relevant, realistic, and clear. Include one open-text field: "What's one email you'll look at differently now?" Track completion-survey
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	response rate as a secondary engagement signal.
Level 2: Learning	Final assessment score (built into the module, 80% pass threshold) and scenario-activity accuracy (percentage of the 4 inbox emails correctly classified) serve as the primary learning measures. Compare average scores across cohorts/departments to identify groups needing reinforcement.
Level 3: Behaviour	Run a simulated phishing campaign (via the organisation's existing phishing-simulation tool, e.g., KnowBe4 or similar) 30–60 days after training, and compare click-through and report rates for trained vs. untrained employees, and pre- vs. post-training for the same cohort. Track how many suspicious emails are forwarded to the security reporting address before vs. after rollout.
Level 4: Results	Track organisation-level security incident metrics over 2–3 quarters following rollout: number of credential-compromise incidents traced to phishing, IT Security team's time spent on phishing-related tickets, and — if available — insurance/audit findings related to phishing risk. A meaningful reduction here is the ultimate business justification for the training investment tied back to the original business problem in Section 1.

Note for portfolio presentation: since this is a self-directed portfolio project without a live corporate rollout, Levels 3 and 4 are presented as the evaluation plan an organisation would execute — this demonstrates the ability to design evaluation strategy beyond the module itself, which is a key differentiator for instructional design roles.

Portfolio Reflection

Suggested framing for a portfolio case study write-up alongside the built module:

- Lead with the business problem and the ADDIE process, not just the finished screens — this shows design thinking, not just software skill.
- Highlight the deliberate interaction choices made to stay realistic for a trial-version, beginner build (e.g., choosing states and layers over custom variables) as an example of scoping a project to real-world constraints.

- Include 3–4 annotated screenshots of the built module: the inbox scenario, the anatomy hotspot slide, and one graded question slide with feedback visible.
- Close the case study with the Kirkpatrick evaluation plan from Section 5 — it demonstrates thinking beyond the module itself, which is often what separates entry-level portfolios from stronger ones.

End of Case Study — Think Before You Click: Spotting Phishing Emails | Portfolio Project